Homework for Chapter 6

1. Find two importance function f_1 and f_2 that are supported on $(1, \infty)$ and are "close" to

$$g(x) = \frac{x^2}{\sqrt{2\pi}} e^{-x^2/2}, \quad x > 1$$

Which of your two importance functions should produce the smaller variance in estimating

$$\int_1^\infty \frac{x^2}{\sqrt{2\pi}} e^{-x^2/2} dx$$

by importance sampling? Explain.

2. We want to compute the following integral by Monte Carlo:

$$\int_0^1 e^{-x\cos(\pi x)} dx = E[h(U)],$$

where U is a uniform distribution on [0, 1] and $h(x) = e^{-x\cos(\pi x)}$. Suppose we use the control variate $Y = g(U)$ with $g(x) = e^{-x}$. Note that we can compute the mean of Y explicitly. By comparing a Monte Carlo estimator with and without control variate, please find the variance reduction from the use of control variate.

3. Estimate the integral $\theta = \int_0^1 e^{x^2} dx$ using Monte Carlo method with $n = 10000$ for the following estimators:

   (a) Regular Monte Carlo estimator.

   (b) Antithetic variable estimator $\frac{1}{2}e^{U^2} + \frac{1}{2}e^{(1-U)^2}$.

   (c) Control variate estimator using U as a control variate.

   (d) Combining the antithetic variable and control variate methods:

$$\hat{\theta}_{\alpha,c} = \alpha e^{U^2} + (1-\alpha)e^{(1-U)^2} + c\left(U - \frac{1}{2}\right).$$

   Try to compute the optimal pair (α^*, c^*) which achieves the smallest variance. Discuss the efficiency of four estimators.

4. Use importance sampling to estimate the quantity:

$$\theta = \int_0^\infty x \frac{e^{-(0.5-x)^2/2} e^{-3x}}{C} dx,$$

where $C = \int_0^\infty e^{-(0.5-x)^2/2} e^{-3x} dx$ is a normalizing constant of a PDF. Plot the converge of the estimator versus sample size. Note: You may consider $3e^{-3x}$ as the importance function. Hint: use self normalized importance sampling.